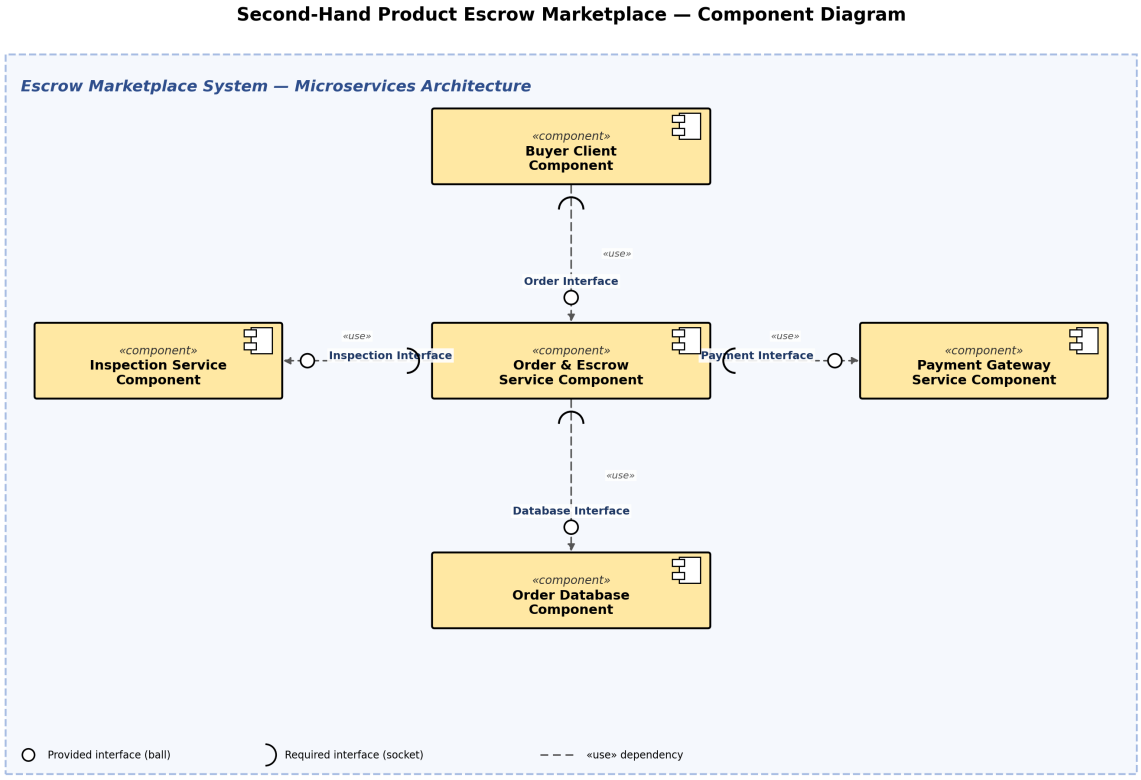
SE LAB-03

# SE LAB - 03

NAME: K CHARITH SAI REDDY
SRN: PES2UG24CS229

## Component Diagram

The diagram below shows five components of the escrow marketplace system, grouped as independent microservices, with provided/required interfaces (ball-and-socket notation) and <<use>> dependencies between them



# Written Justification

## Architecture Selection

We chose Microservices Architecture for the Second-Hand Product Escrow Marketplace System.

## Reasons

->The marketplace has clearly separable domains — order/escrow management, payment processing, and the inspection workflow — each with different scaling and reliability needs (payment traffic spikes differently from inspection traffic). Splitting them into independently deployable services lets each one scale and be updated on its own schedule without redeploying the whole system.

->Escrow funds and inspection reports must stay available and consistent even if one part of the system misbehaves. Fault isolation means a slow or failing Inspection Service cannot take down order browsing or payment processing — which matters for a marketplace handling real money.

## Security Advantage

Payment handling is isolated inside its own Payment Gateway Service Component, reachable only through a defined Payment Interface from the Order & Escrow Service. Card and payment-gateway credentials never need to touch the Buyer Client or the Inspection Service, narrowing the attack surface and making it easier to apply stricter access control and auditing to that one service.

## Performance Benefit

Because each service can be scaled independently, the system can add more Order & Escrow Service or Payment Gateway Service instances during high-traffic periods (e.g. sale events) without over-provisioning the Inspection Service — keeping checkout and escrow operations responsive even under load.